The engineers who revolutionised Formula 1

From Cooper to Newey: who really changed the technical grammar of the sport?

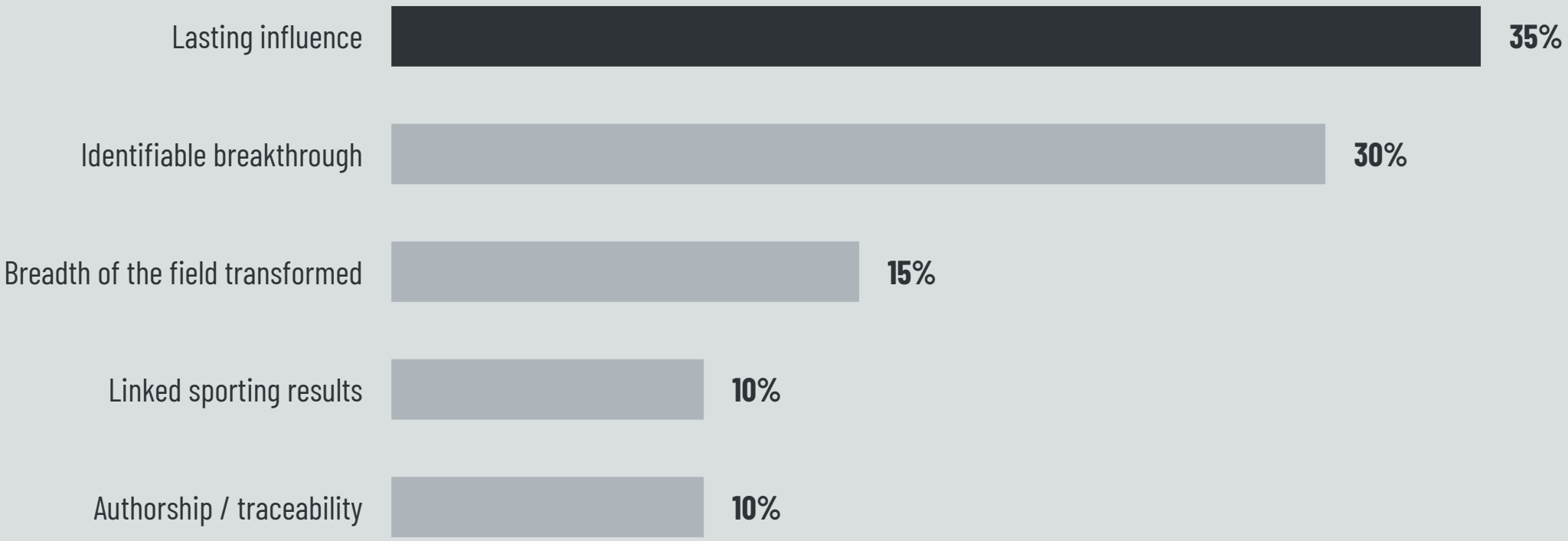


“Who won the most?” is not the right question

Titles	≠	**Innovation**
Innovation	≠	**Invention**
Invention	≠	**Diffusion**
Technical direction	≠	**Personal authorship**

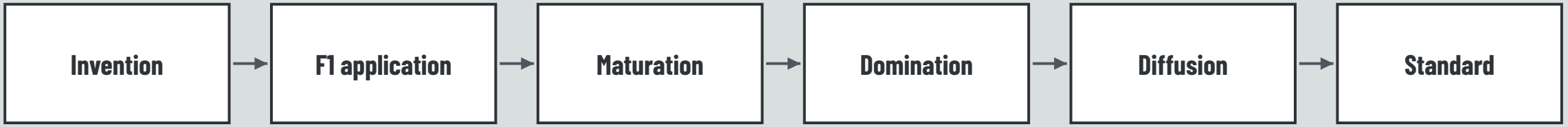
The ranking measures the capacity to shift the technical frontier for good.

Five criteria, led by lasting influence



The scores are a comparative instrument, not historical facts.

An innovation has several authors, at different stages



Examples

Ground effect	Cooper rear engine	Cosworth DFV	Williams active suspension

Twenty engineers, four tiers

Tier A
Paradigm revolutionaries


Colin Chapman


John Barnard

JC John Cooper

KD Keith Duckworth


Adrian Newey

Tier B
Major systemic transformers

BD Bernard Dudot


Gordon Murray


Mauro Forghieri

PW Peter Wright

FD Frank Dernie

Tier C
Structuring innovators

PH Patrick Head


Ross Brawn

AC Andy Cowell

JM Jean-Claude Migeot


Tony Rudd

Tier D
More localised or more shared impact


Mike Costin


Vittorio Jano

OM Owen Maddock


Paddy Lowe


Rory Byrne

Photos: credits on slide 22

Five questions that changed F1

1	JC	**Cooper**	Where should the engine go?
2		**Chapman**	How should the car be built?
3	KD	**Duckworth**	How should the engine be integrated?
4		**Barnard**	With which materials and interfaces?
5		**Newey**	How can the whole aerodynamic system be optimised?

Photos: credits on slide 22

Colin Chapman

Three breakthroughs, one philosophy: architecture + lightness + aero

Lotus 25

Aluminium monocoque

Lotus 49

Engine as a stressed member

Lotus 78 / 79

Ground effect programme



Shared attribution: Wright / Rudd for ground effect.

Rank	Tier	Score	Confidence
1	A	96.0	High

Sources: Lotus — Type 25 · Lotus — Lotus Story · F1 — Tech Tuesday: The Lotus 79 · Photo: Joost Evers / Anefo (Nationaal Archief) (CC0, Wikimedia Commons), cropped and converted to black and white

John Barnard

Two independent innovations that became universal

1981

McLaren MP4/1

Full carbon-fibre monocoque

1989

Ferrari 640

Semi-automatic gearbox, paddles on the wheel



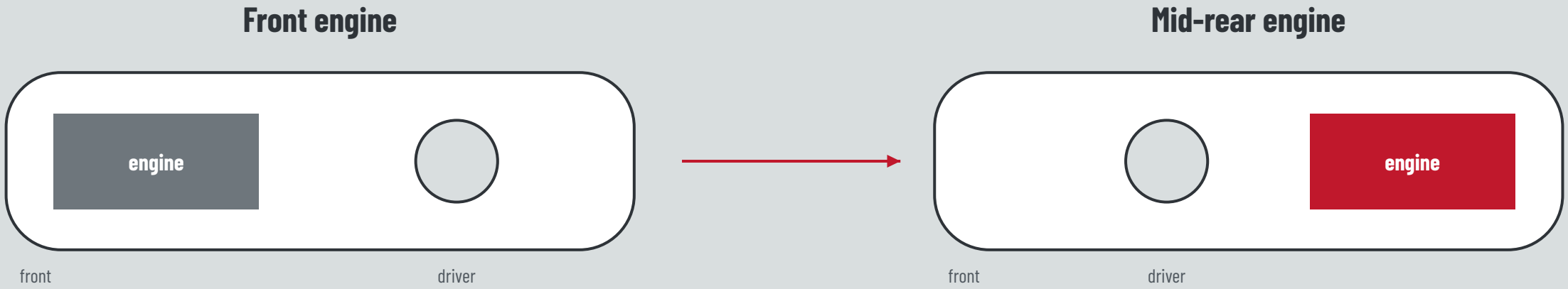
The clearest personal authorship: Chapman's main challenger.

Rank	Tier	Score	Confidence
2	A	95.0	Very high

Sources: McLaren — MP4/1 · F1 — John Barnard podcast · F1 — How F1 looked in 1990 · Photo: B3 Technologies (CC BY-SA 3.0, Wikimedia Commons), cropped and converted to black and white

John Cooper

A topological revolution of the single-seater



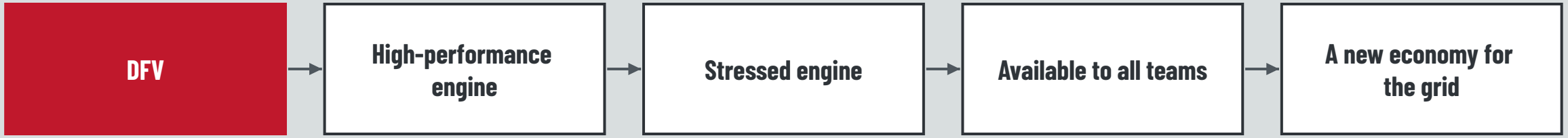
Cooper T51, 1959 and 1960 titles
Chief designer: Owen Maddock

Rank	Tier	Score	Confidence
3	A	93.5	High

Sources: Motor Sport — Cooper T51 · Goodwood / Doug Nye — Cooper · F1 — Most influential technical innovator

Keith Duckworth

A technical and economic revolution at once



155 world championship Grand Prix wins, 1967-1983

Cosworth = Duckworth + Costin + an organisation

Rank	Tier	Score	Confidence
4	A	92.5	High

Sources: F1 — Most influential technical innovator · Motor Sport — The Lotus 49 (1967)

Newey's genius is repetition, not a single stroke

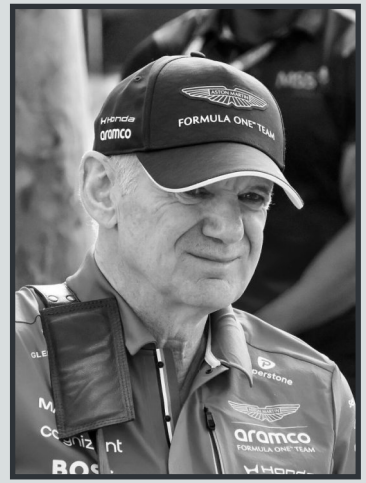
March / Leyton House — 1988-1990

Williams — 1991-1996

McLaren — 1997-2005

Red Bull — 2006-2024

Aston Martin — since 2025



Different regulations, the same skill: mastery of the aerodynamic system.

Rank	Tier	Score	Confidence
5	A	91.5	Very high

Sources: F1 — Most influential technical innovator · F1 — Newey's Red Bull exit · F1 — How F1 looked in 1990 · Photo: Yu Chu Chin (CC BY-SA 4.0, Wikimedia Commons), cropped and converted to black and white

The real debate for first place

Why Chapman ahead of Barnard?



Criterion	Chapman	Barnard
Number of breakthroughs	++	+
Direct authorship	+	++
Architectural reach	++	++
Diffusion	++	++
Repetition	++	+



Chapman wins on historical breadth; Barnard wins on personal traceability.

Photo: Joost Evers / Anefo (Nationaal Archief) (CC0, Wikimedia Commons), cropped and converted to black and white · Photo: B3 Technologies (CC BY-SA 3.0, Wikimedia Commons), cropped and converted to black and white

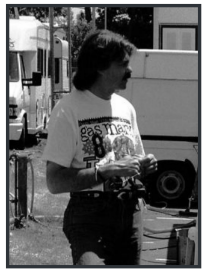
Tier B: five systemic transformers

BD

6

Dudot

Power



7

Murray

Radical architecture



8

Forghieri

Total engineering

PW

9

Wright

Ground effect

FD

10

Dernie

A silent digital revolution

Compared with Tier A: a more collective attribution, a more specialised reach or uneven diffusion.

Sources: Renault — 30 ans de Formule 1 · F1 — Most Influential Person: final 4 · F1 — Ferrari 312T · F1 — Tech Tuesday: The Lotus 79 · Motor Sport — Lunch with Frank Dernie · Photos: credits on slide 22

The engine revolution: turbo, pneumatic valves, V10



Adopted by the whole grid

The general solution in the 1990s

67° V10 under Dudot's direction

Renault Sport is an organisation: Dudot is not a lone inventor.

Rank	Tier	Score	Confidence
6	B	90.0	High

Sources: Renault — 30 ans de Formule 1 · Motor Sport — Renault Returns

The silent revolutions

Frank Dernie

- CAD / CAE
- Data acquisition
- Wind tunnel
- Active suspension

FD

Ross Brawn

- Methods
- Organisation
- Early programmes
- Integration



A technical revolution can transform the process without showing on the bodywork.

Sources: Motor Sport — Lunch with Frank Dernie · F1 — Ross Brawn favourite car designs · Photos: credits on slide 22

Ground effect: restoring the causal chain



Lotus 78: drawn by Ralph Bellamy, suspension by Martin Ogilvie, development with Tony Southgate.

Sources: F1 — Tech Tuesday: The Lotus 79

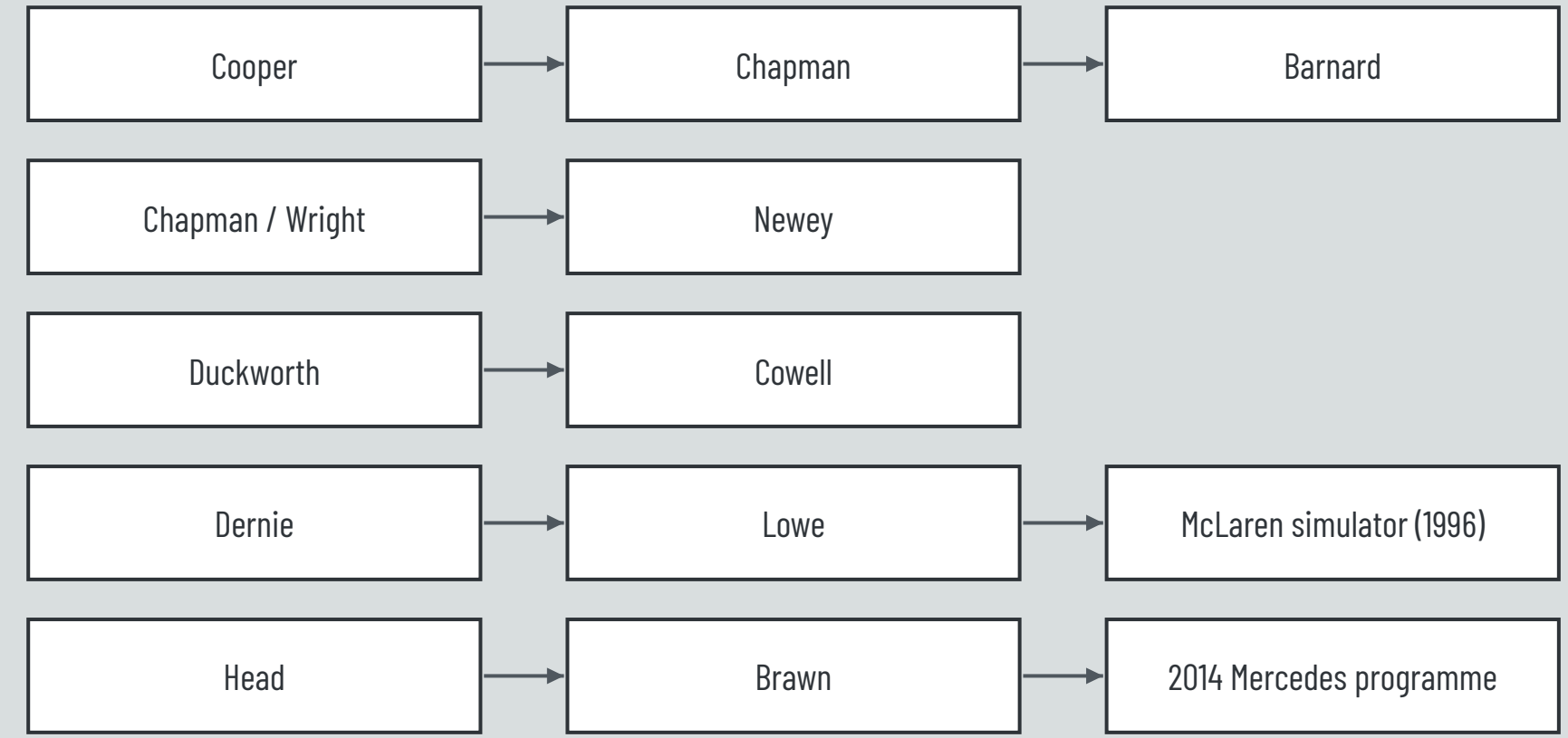
Ranks 11 to 20

Tiers C and D: ten major contributions

Rank		Engineer	Field	Reason for the ranking	Limit
11	PH	**Patrick Head**	Systems engineering	Williams technical organisation	Organisation rather than a single breakthrough
12		**Ross Brawn**	Method	Early development, integration	A revolution in process
13	AC	**Andy Cowell**	Power unit	Mercedes 2014 hybrid	Hybridisation imposed by the regulations
14	JM	**Jean-Claude Migeot**	Aerodynamics	Raised nose, Tyrrell 019	Quickly absorbed by general evolution
15		**Tony Rudd**	Aerodynamics	Origins of ground effect	Contribution hard to isolate
16		**Mike Costin**	Engine	Co-founder of Cosworth	DFV authorship less individual
17		**Vittorio Jano**	Architecture	Lancia D50	Work mostly before F1
18	OM	**Owen Maddock**	Chassis	Chief designer, 1959-1960 titles	Credit shared with John Cooper
19		**Paddy Lowe**	Electronics	Active suspension software	A role in a collective chain
20		**Rory Byrne**	Integration	Benetton, Ferrari	No single paradigmatic breakthrough

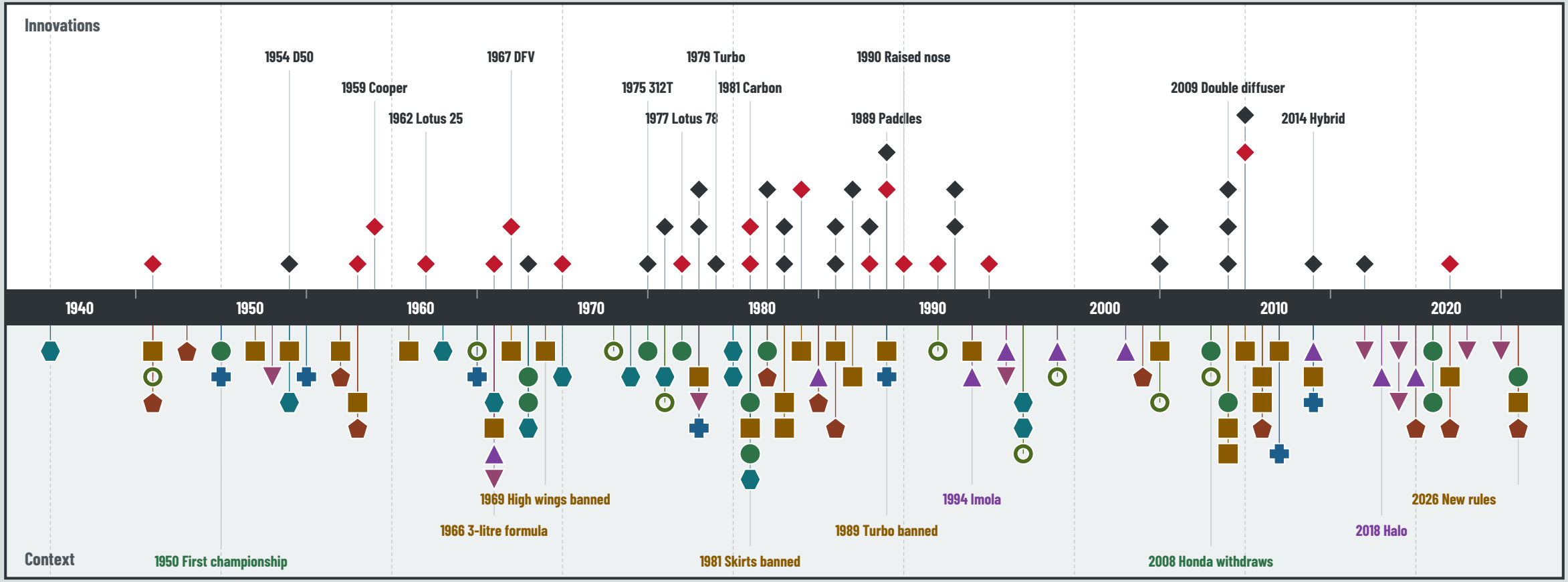
Sources: F1 — Williams FW15C · F1 — Ross Brawn favourite car designs · F1 — How F1 looked in 1990 · Stellantis Heritage — Lancia D50 · Motor Sport — Cooper T51 · Photos: credits on slide 22

The great technical lineages



This diagram shows intellectual lineages, not master-pupil relationships.

Eighty years of breakthroughs, and their context

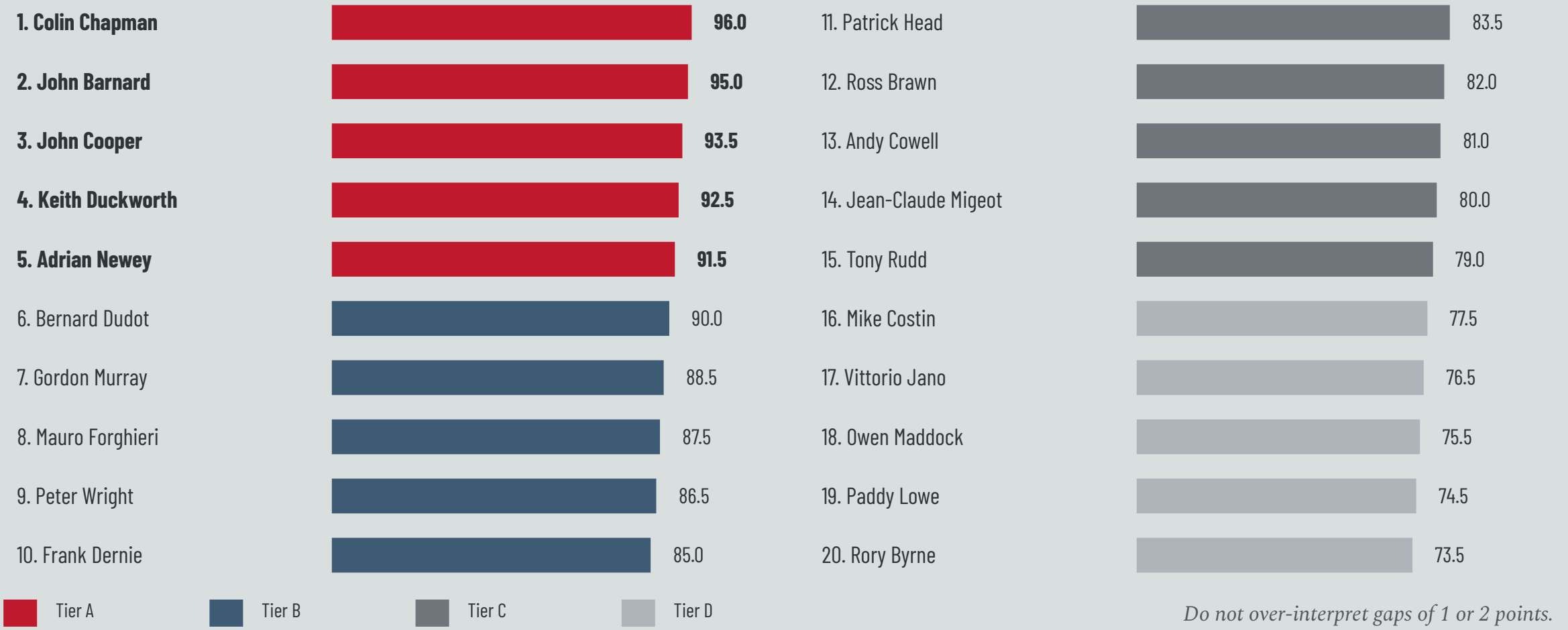


innovation, Tier A · innovation · Regulations · Safety · Economics · Technology · Geopolitics · World economy · Other sports · Society

Interactive timeline: s-geffroy.github.io/f1-eng/en/timeline

Scores out of 100

The final ranking



Do not over-interpret gaps of 1 or 2 points.

Revolutionising F1 is not just about building the fastest car: it is about forcing others to change the way they build an F1 car.

Chapman	*multiple breakthroughs*
Barnard	*the clearest authorship*
Newey	*the most durable adaptation*

Consulted on 9 August 2026

Main sources

- F1 — Most influential technical innovator
- F1 — The 75 best… in F1 history
- F1 — Tech Tuesday: The Lotus 79
- F1 — John Barnard podcast
- F1 — Williams FW15C
- Lotus — Type 25
- McLaren — MP4/1
- Renault — 30 ans de Formule 1
- Motor Sport — Cooper T51
- Motor Sport — Lunch with Frank Dernie

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Full bibliography, with addresses: speaker notes (.pptx) and s-geffroy.github.io/f1-eng/en